Md Abdul Ahad Juel

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Summary

A dedicated electrical engineer with over 8 years of experience in renewable energy systems, power electronics integration, and industrial automation, seeking a PhD position to advance research in sustainable power systems, grid integration of renewable energy, and smart energy storage solutions. Proven ability to lead large-scale engineering projects, conduct system-level analysis, and contribute to interdisciplinary research aimed at enhancing the efficiency, reliability, and resilience of modern electrical grids. Eager to apply academic and industrial expertise to solve pressing energy challenges through innovative research.

Skills

Technical: Control Systems, Substation Design & Engineering, Power Distribution, Renewable Energy Integration, Electrical Equipment Maintenance, Transformer & Switchgear O&M, Power Quality Analysis.

Project Management: Project Planning & Execution, Budget Management, Resource Allocation, Risk Assessment, Schedule Management, Vendor Management.

Compliance & Safety: NFPA, NEC, OSHA, WHMIS, Confined Space, Electrical Safety Awareness, Regulatory Compliance, Permitting.

Tools & Software: MS Office, AutoCAD, Data Analysis & Reporting Systems.

Education

Master of Science – Electrical and Electronics Engineering *Present*
University at Buffalo, SUNY

Master of Arts – Community Economic Development
Western Illinois University

Bachelor of Science – Electrical and Electronics Engineering
Bangladesh University of Business & Technology (BUBT)

Google Scholar: *https://scholar.google.com/citations?hl=en&user=vsbOZiwAAAAJ*
ORCID ID: *0009-0002-2578-0325*

Professional Experience

Senior Executive Engineer *Feb 2017 – Dec 2021*

Super Star Renewable Energy Limited | Bangladesh

- Led installation and commissioning of over 2,500 solar and power systems projects, from residential to industrial scale, delivering 27.19 MWp of total installed capacity.
- Designed and executed a 2 MWp solar grid-tied system for Pacific Jeans Ltd., fulfilling 20% of their total electrical energy demand using smart inverter technology.
- Completed installation and commissioning of a 250 kWp solar mini-grid for a remote island, meeting the energy needs of 1,200 households.
- Integrated electrical components including inverters, batteries, combiner boxes, CDPs, and transformers to ensure seamless system operation per design specifications.
- Installed PV systems across residential, commercial, and industrial environments including rooftop, off-grid, on-grid, and hybrid configurations.
- Deployed solar modules, inverters, racking systems, and ancillary electrical apparatus across diverse project sites.
- Engaged clients to define project objectives, ensuring full alignment with technical and operational requirements.
- Implemented electrical wiring, transformers, and switchgear to support solar power infrastructure.

Engineer – Industrial Instrumentation & Automation *Jan 2015 – Nov 2016*

Industrial Instrument and Automation Solution (IIAS) | Bangladesh

- Identified and specified appropriate sensors, transducers, and measurement instruments for industrial processes including temperature, pressure, flow, and level monitoring.
- Ensured seamless integration of new instrumentation and automation solutions with existing machinery and software platforms.
- Implemented real-time energy monitoring solutions to track usage and identify conservation opportunities.
- Developed automation workflows to optimize processes for minimal energy consumption and reduced environmental impact.
- Designed autonomous robotic systems for precision assembly, inspection, and packaging in hazardous or high-accuracy environments.

Sub-Assistant Engineer – Electrical *Mar 2013 – May 2014*

Powerbreeze Engineering Ltd. | Bangladesh

- Monitored performance of electrical systems including generators, transformers, and control equipment, ensuring optimal operation.
- Developed and implemented preventive maintenance schedules; conducted routine inspections to minimize downtime and ensure equipment reliability.
- Diagnosed and resolved electrical faults through root cause analysis and corrective action implementation.
- Enforced safety protocols, delivered safety training, and conducted regular audits to maintain compliance with standards and regulations.
- Identified energy efficiency opportunities; analyzed usage data and implemented measures to reduce waste.
- Led renewable energy integration efforts including solar, wind, and hydroelectric power, optimizing output and sustainability.
- Managed electrical system upgrade projects encompassing planning, budgeting, scheduling, and stakeholder coordination.
- Analyzed operational data, monitored performance metrics, and produced management and regulatory reports.
- Coordinated emergency response to equipment failures and power outages, restoring power safely and efficiently.

Licenses & Certifications

- e-Electrical Supervisor's Competency Certificate – Ministry of Power, Energy & Mineral Resources, Bangladesh
- Electric Power Systems – University at Buffalo & SUNY, USA
- Renewable Energy Technology Fundamentals – University of Colorado Boulder, USA
- Electrical Power Generation: An Industrial Outlook – L&T EduTech
- Engineering Management: Scratch to Expert – Udemy
- GE Aerospace – Explore Engineering Job Simulation – Forage
- GE Aerospace – Explore Supply Chain Job Simulation – Forage
- Health & Safety: Confined Space Hazard Awareness for Construction – IHSA
- Health & Safety: Electrical Safety Awareness – IHSA
- WHMIS Certificate – Infrastructure Health & Safety Association (IHSA)
- LEAN SIX SIGMA WHITE BELT-The Council for Six Sigma Certification

Training

- Lab Safety Training (In-Person) – University at Buffalo.
- National e-Government Procurement Training – Training Authority, Bangladesh.
- Solar Street Light Installation & Commissioning.
- Solar PV System Installation and Commissioning.
- Solar Mini-Grid Operation and Maintenance.